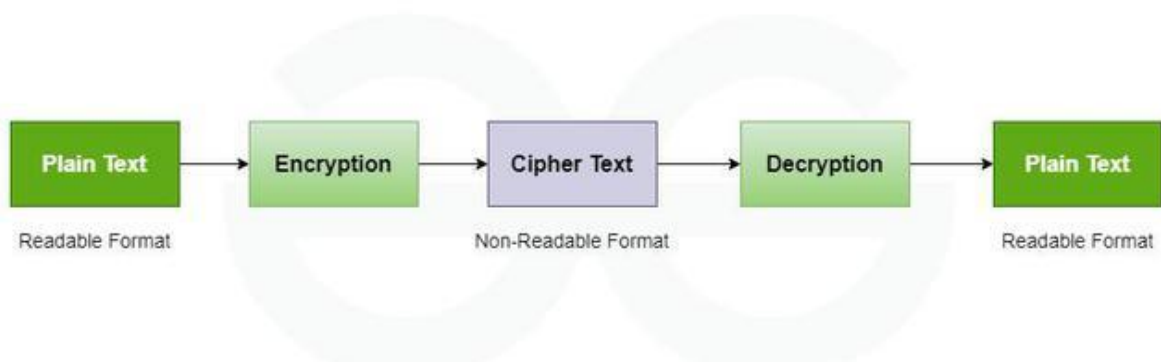


Unit 3 Cyber Threats and Attack Techniques

3.1 Cryptography:

Cryptography is the science of protecting information using mathematical techniques to ensure confidentiality, integrity, and authentication. It transforms readable data into unreadable form, preventing unauthorized access and tampering.

- Converts plaintext into ciphertext using algorithms and keys
 - Ensures confidentiality, integrity, authentication, and non-repudiation
 - Used in secure communication, digital signatures, passwords, and online transactions
-
- **Confidentiality:** Only the intended recipient can read the message.
 - **Integrity:** The message hasn't been altered in transit.
 - **Authentication:** You can prove who sent the message (or that it came from a legitimate source).
 - **Non-repudiation:** The sender cannot later deny having sent it.



These algorithms are used for cryptographic key generation, digital signing, and verification to protect data privacy, web browsing on the internet and to protect confidential transactions such as credit card and debit card transactions.

1. Plain text and Ciphertext:

Plaintext: Original readable message

Ciphertext: Encrypted, unreadable version

2. Encryption and Decryption

Encryption: Converting plaintext to ciphertext

Decryption: Converting ciphertext to plaintext.

3. Keys: A key is secret value used in encryption and decryption. Security depends heavily on keeping the key safe.

Unit 3 Cyber Threats and Attack Techniques

Common Cryptographic Techniques

1. Symmetric Key Cryptography

- ✓ Same key used for encryption and decryption
- ✓ Fast and efficient
- ✓ Examples: AES (Advanced Encryption Standard)

Used in: file encryption, disk encryption

2. Asymmetric Key Cryptography

- ✓ Uses a pair of keys:
 - **Public key** (shared)
 - **Private key** (kept secret)
- ✓ Slower but more secure for key exchange

Used in: secure communication (HTTPS), email encryption

3. Hash Functions

- ✓ Convert data into a fixed-size hash value
- ✓ One-way function (cannot reverse easily)

Used in: password storage, data integrity checks

4. Digital Signatures

- ✓ Provide authenticity and integrity
- ✓ Created using private key, verified with public key

Used in: software distribution, online transactions

Application of Cryptography:

- ✓ Secure web browsing (HTTPS)
- ✓ Online banking and payments
- ✓ Cryptocurrencies like Bitcoin
- ✓ Password protection systems
- ✓ Military and government communications

Types of Cryptography

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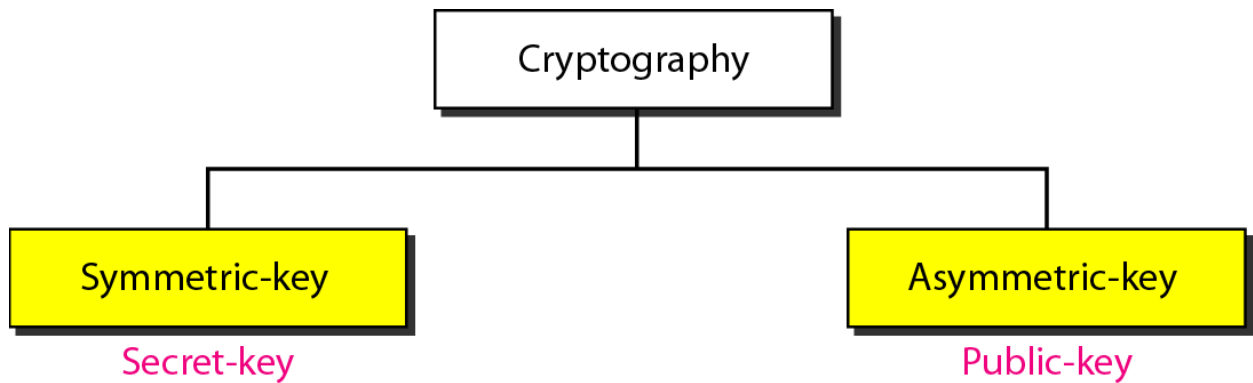


Figure: Types of Cryptography

Symmetric Key Cryptography

Symmetric encryption (also called classical cryptosystems), also referred to as conventional encryption, secret key, or single-key encryption, was the only type of encryption in use prior to the development of public-key encryption in the late 1970s.

Same key is shared by sender (for encryption) and receiver (for decryption). - Ciphertext is generated either by substitution or transformation method by encryption algorithm.

A Symmetric encryption has five ingredients:

- Plaintext
- Encryption Algorithm
- Secret Key
- Ciphertext
- Decryption Algorithm

In cryptography, **ciphers** are algorithms used to convert readable data (**plaintext**) into unreadable form (**ciphertext**) so that unauthorized people cannot understand it. Two important types of symmetric encryption methods are:

- **Stream cipher**
- **Block cipher**

Both are used to protect data, but they work in different ways.

Stream Cipher:

A **stream cipher** encrypts data **one bit or one byte at a time**.

Instead of taking a large chunk of data at once, it processes the data as a continuous stream. That is why it is called a **stream cipher**.

Basic idea

A stream cipher generates a **keystream** (a sequence of random-looking bits) using a secret key. This keystream is then combined with the plaintext bit by bit or byte by byte, usually using the **XOR operation**.

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Ciphertext = Plaintext XOR Keystream

To get the original message back:

Plaintext = Ciphertext XOR Keystream

Because XOR with the same keystream reverses the operation.

Example of Stream Cipher

Suppose:

- Plaintext = 10110010
- Keystream = 11001001

Now apply XOR:

- 10110010
- 11001001
- Result = 01111011

So the ciphertext becomes 01111011.

To decrypt, XOR again with the same keystream:

- 01111011
- 11001001
- Result = 10110010

Original plaintext is recovered.

Characteristics of Stream Cipher

1. Encrypts data continuously

It works on data as it comes, one unit at a time.

2. Fast

Stream ciphers are usually very fast and efficient.

3. Suitable for real-time communication

They are commonly used when data is flowing continuously, such as:

- voice calls
- video streaming
- wireless communication

4. No padding required

Since data is handled bit by bit or byte by byte, there is no need to fill extra space.

Advantage of Stream Cipher

- Very fast in execution
- Requires less memory
- Good for live or continuous data transmission
- No padding problem

Disadvantages of Stream Cipher

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- If the **same keystream** is reused, security becomes very weak
- Small implementation mistakes can lead to serious vulnerabilities
- Usually less suitable for file-based encryption than block ciphers

Block Cipher: A **block cipher** encrypts data in **fixed-size blocks** instead of processing it one bit or byte at a time.

For example, a block cipher may take:

- 64 bits at a time, or
- 128 bits at a time

Each block of plaintext is transformed into a block of ciphertext using a secret key.

Basic idea

If the plaintext is larger than one block, it is divided into multiple blocks, and each block is encrypted.

For example, if a cipher uses a **128-bit block size**, then the plaintext is broken into 128-bit pieces.

Example of Block Cipher

Suppose a block cipher works on **8-character blocks**.

Plaintext message:

COMPUTERSECURITY

This may be divided into blocks like:

- COMPUTER
- SECURITY

Each block is encrypted separately using the same key (depending on the mode of operation).

Characteristics of Block Cipher

1. Fixed-size processing

It works on blocks of fixed size.

2. More structured encryption

Since it works on chunks, it is commonly used for stored data such as files and databases.

3. Requires modes of operation

A block cipher by itself only encrypts one block at a time. To securely encrypt long messages, special methods called **modes of operation** are used, such as:

- ECB
- CBC
- CFB

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- OFB
- CTR
- GCM

Advantages of Block Cipher

- Strong and widely trusted
- Good for encrypting files, disks, and database records
- Can provide both confidentiality and, with some modes, integrity
- Very common in practical security systems

Disadvantages of Block Cipher

- Usually slower than stream ciphers for real-time bit-by-bit encryption
- Padding may be needed if plaintext size is not equal to block size
- Poor mode selection can create vulnerabilities

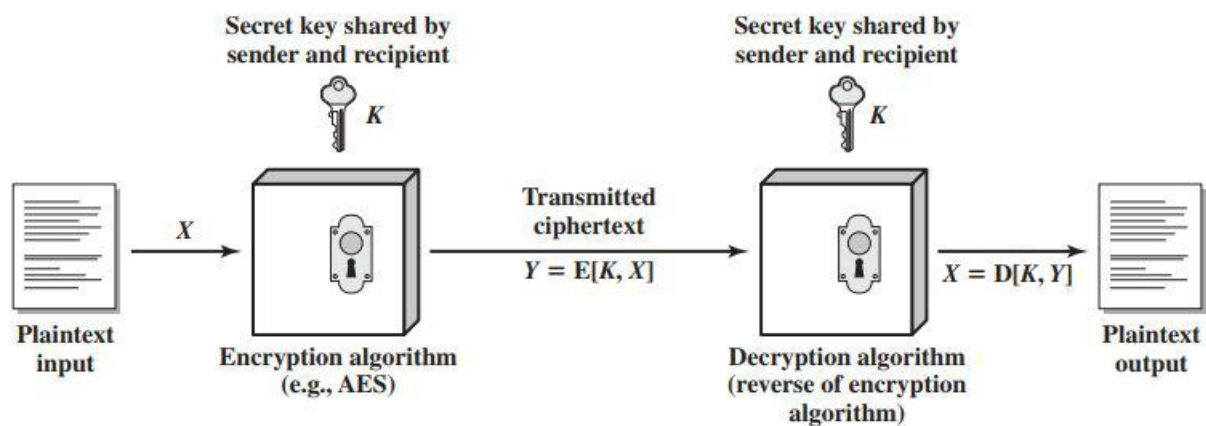


Figure: Simplified Model of Symmetric Encryption

There are two requirements for secure use of symmetric encryption:

1. We need a strong encryption algorithm.
 - At minimum, an operator who knows the algorithm and has access to one or more ciphertext would be unable to decipher the ciphertext.
2. Sender and receiver must have obtained copies of the secret key in a secure fashion and must keep the key secure.

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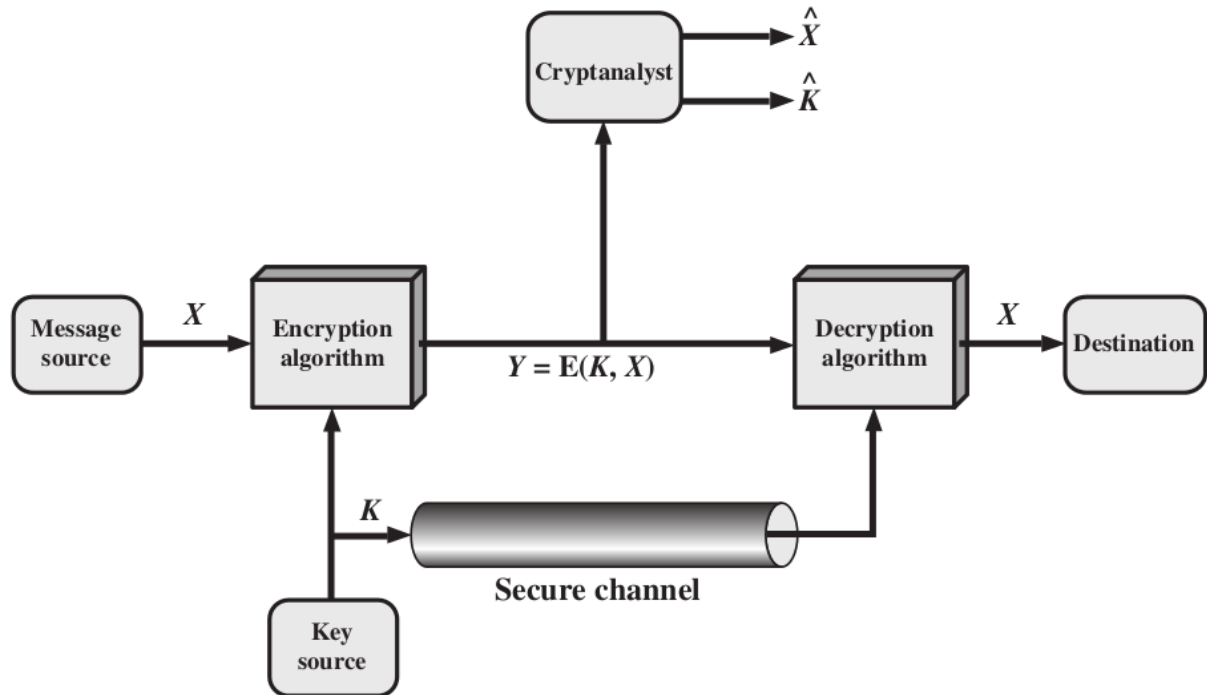


Figure: Model of symmetric key cryptography

There are two basic types of classical ciphers:

- i. Transposition ciphers
- ii. Substitution Ciphers

i. Transposition Ciphers

- ✓ A transposition cipher does not substitute one symbol for another; instead, it changes the location of the symbols.
- ✓ A symbol in the first position of the plaintext may appear in the tenth position of the ciphertext.
- ✓ A symbol in the eighth position in the plaintext may appear in the first position of the ciphertext.
- ✓ In other words, a transposition cipher reorders (transposes) the symbols.

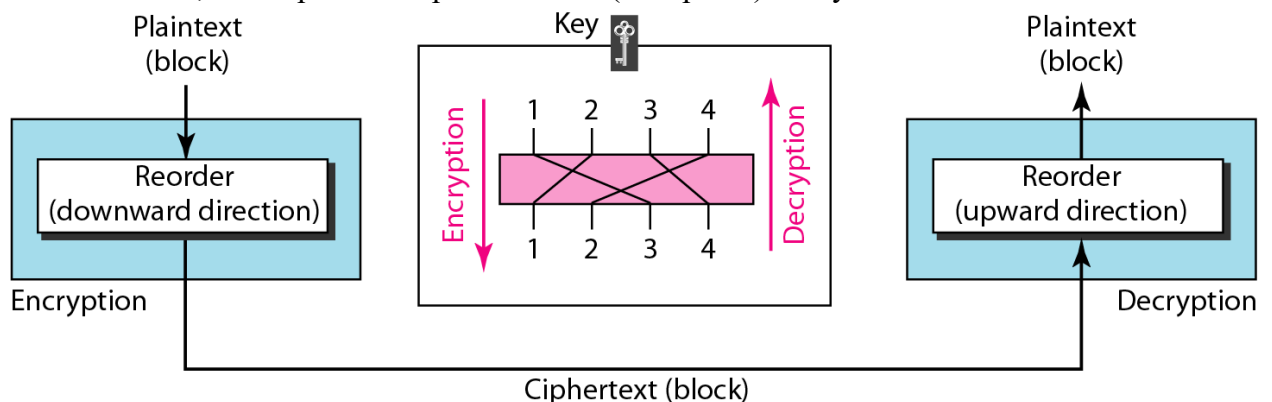


Figure: Transposition Cipher

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For Example: Suppose Alice wants to secretly send the message "Enemy attacks tonight" to Bob. The Encryption and decryption is shown in Figure. Note that we added an extra character(z) to the end of the message to make the number of characters a multiple of 5.

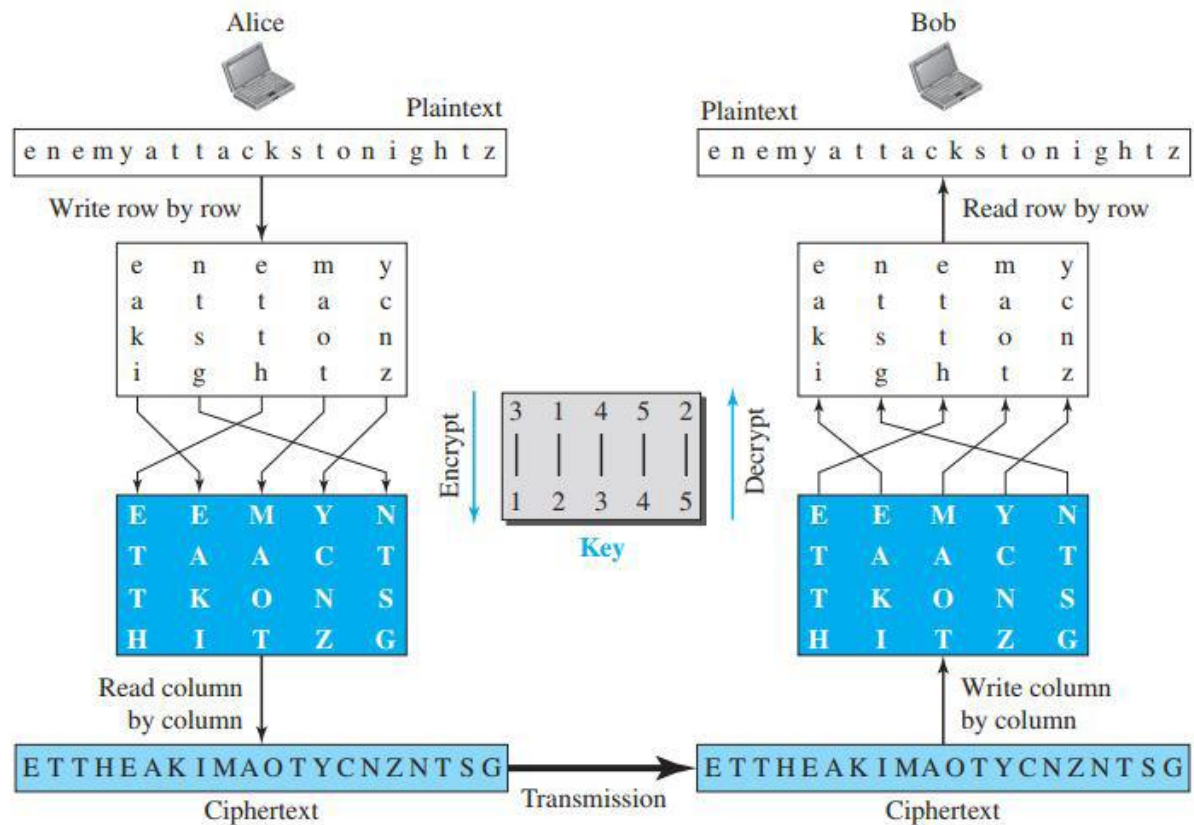


Figure: An example of Substitution cipher

ii. Substitution Cipher:

- A substitution cipher changes characters in the plaintext to produce the ciphertext.
- For example: Caesar cipher, Vigenere Cipher, One-time pad are example of substitution cipher.

Caesar Cipher

- Substitution cipher
- Shift cipher
- Earliest known and simplest substitution scheme developed by Julius Caesar.
- Replaces each letter of the alphabet with the letter standing three places further down the alphabet.

plain: meet me after the toga party

cipher: PHHW PH DIWHU WKH WRJD SDUWB

- A Caesar cipher is susceptible to a statistical ciphertext-only attack.

Cryptanalysis

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Cryptanalysis is the study of analyzing and breaking cryptographic systems. It aims to uncover hidden information without access to the secret key.

- Recover plaintext from ciphertext
- Discover the secret key
- Identify weaknesses in cryptographic algorithms

Classical Methods of Cryptanalysis

1. Frequency Analysis

- Studies how often letters appear
 - Effective against simple substitution ciphers
- Example: In English, 'E' appears most frequently

2. Brute Force Attack

- Try all possible keys until the correct one is found
- Time-consuming but guaranteed if enough time is available

Feistel Cipher Structure

Horst Feistel proposed the concept of a product cipher;

- the execution of two or more simple ciphers in sequence
- final product is cryptographically stronger than any of the component ciphers.

Feistel cipher structure consists of a block cipher with a key length of k bits and a block length of n bits;

- Has to undergo 2^k possible transformations only (2^n transformations in ideal block cipher).

Feistel cipher makes use of substitutions and permutations alternatively : -

Substitution: Each plaintext element or group of elements is uniquely replaced by a corresponding ciphertext element or group of elements.

Permutation: A sequence of plaintext elements is replaced by a permutation of that sequence. (i.e. the order of the elements is changed.)

The Feistel cipher structure, which is based on Shannon's proposal of **Diffusion and Confusion** in 1945, is the structure used by many significant symmetric block ciphers currently in use.

Inputs to the encryption algorithm are a plaintext block of length $2w$ bits and a key K .

The plaintext block is divided into two halves, L_0 and R_0 .

The two halves of the data pass through n rounds of processing and then combine to produce the ciphertext block.

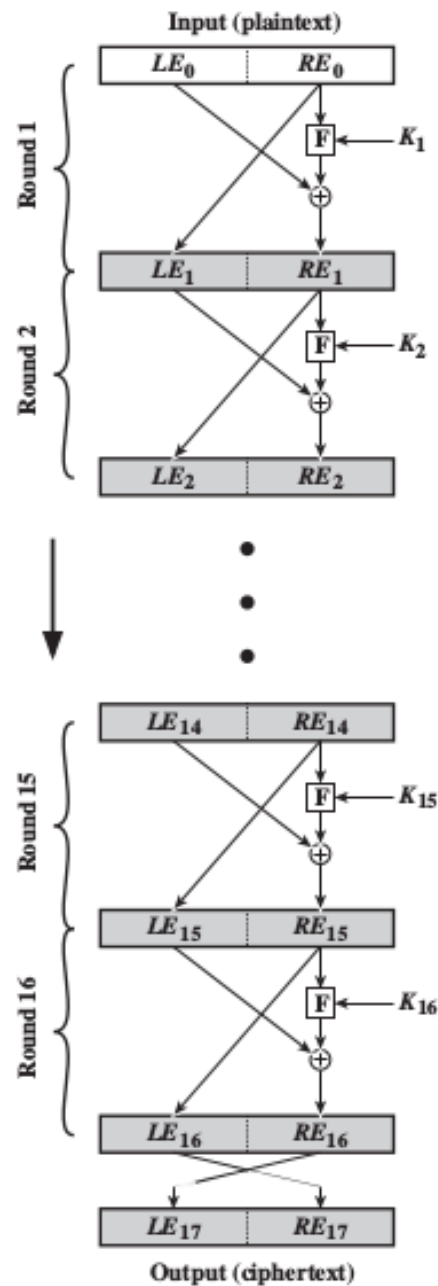
Each round i has as inputs L_{i-1} and R_{i-1} derived from the previous round, as well as a subkey K_i derived from the overall K .

A substitution is performed on the left half of the data.

- Apply a round function F to the right half of the data and then take the X-OR of the output of that function and the left half of the data.

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- F has the same general structure for each round but is parameterized by the round subkey K_i . Following this substitution, a permutation is performed that consists of the interchange of the two halves of the data.

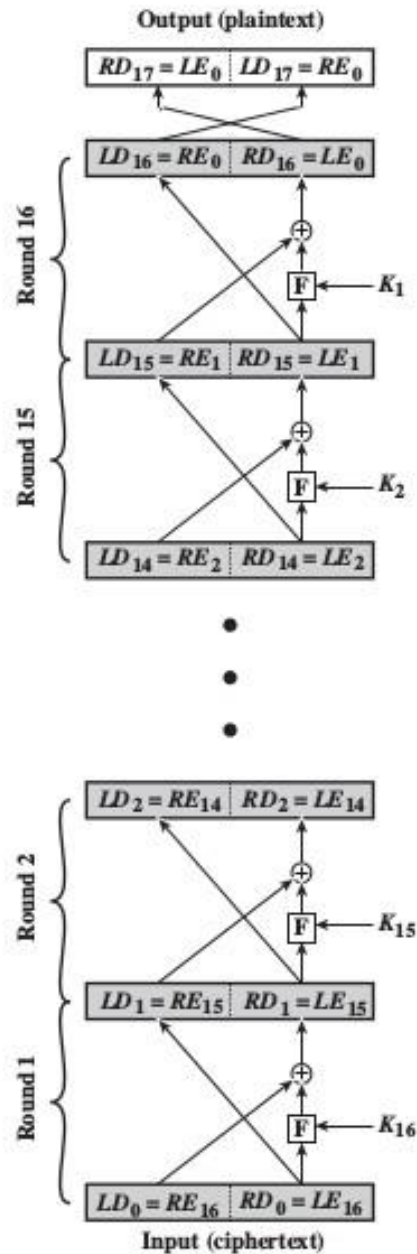


Decryption has same process as the encryption; only the input keys parameterize it.

The subkeys are used in reverse order, i.e. K_{16} is used in the first round of decryption.

The first round of decryption reverses the last round of encryption, the second round of decryption reverses the second last round of encryption, and so on.

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Feistel Cipher Analysis:

The realization of a Feistel network depends on : -

- **Block size:**
 - Larger block sizes mean greater security but reduced speed for algorithm.
- **Key size:**
 - Larger key size means greater security but decreases encryption/ decryption speed.
- **Number of rounds:**
 - Multiple rounds offer increasing security. A typical size is 16 rounds.
- **Subkey generation algorithm:**
 - Greater complexity leads to greater difficulty of cryptanalysis.

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- **Round function F:**

- A complex function means greater resistance to cryptanalysis.

Other considerations : -

Fast software encryption/decryption:

In case of a hardware implementation, the speed of execution of the algorithm becomes a concern.

- **Easy analysis of algorithm:**

- If the algorithm can be concisely and clearly explained, it is easier to analyze that algorithm for cryptanalytic vulnerabilities.

Data Encryption Standard (DES Encryption)

- First developed in 1974 by IBM and NSA (based on LUCIFER that was 64-bit encryption with 128 bit key size).
- Also known as Data Encryption Algorithm (DEA) that was adopted in 1977 by National Bureau of Standards (presently called NIST) as the Data Encryption Standard.
- Data are encrypted in 64-bit blocks using a 56-bit key.
- The same steps, with the same key, are used to reverse the encryption.
- DES has the same structure as Feistel cipher except the F function and additional initial and final permutations IP and IP⁻¹, respectively.

Step1: Input plain text

64 bit plain text as input

Step2: Initial Permutation

Only bit pattern changed.

Step3: Divide two halves

L₀ → 32 bits

R₀ → 32 bits

Step4: Key Preparation

- 64 bit Key
- Permitted Choice 1(PC1)- removes 8 parity bits → 56 bits
- Split into two - 28 bits halves
- Prefer left Circular Shift
- Apply Permitted Choice2 (PC2) to generate 48bits round key
- This is repeated to generate 48bits round key.

Step5: Round of Encryption

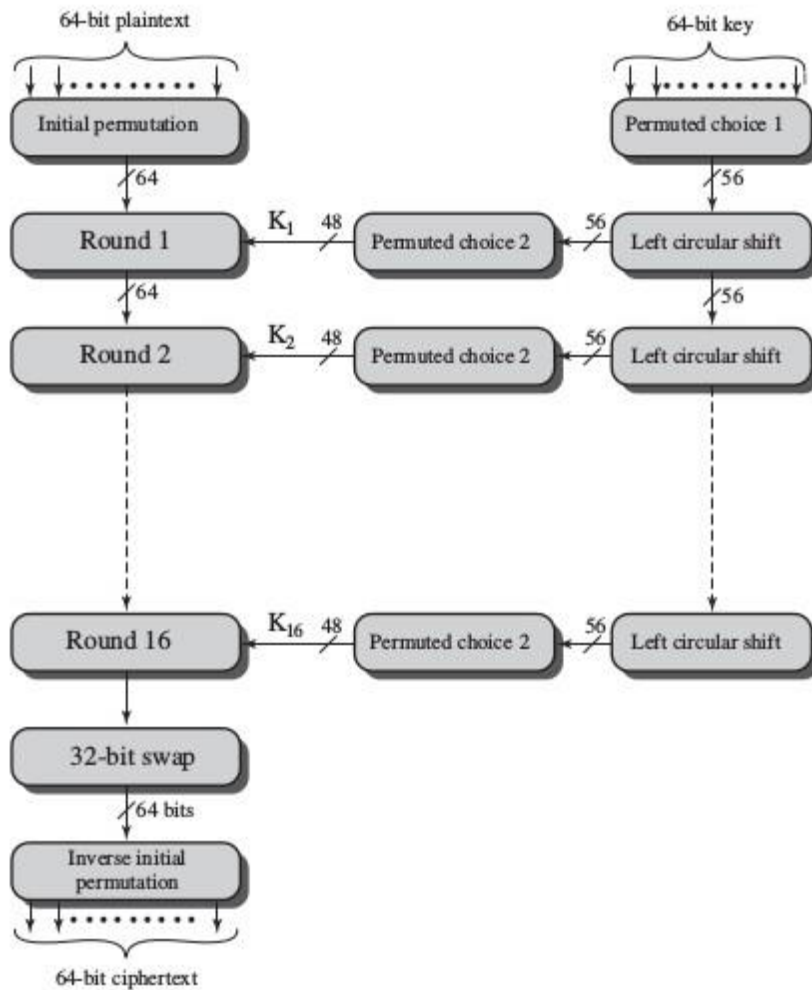
- Expand R_{i-1} (32bits → 48bits) using expansion Permutation
- XOR the expanded R with Round key k_i
- Pass the result through S-Boxes (48bits - 32bits)
- Apply permutation (P-Boxes)
- XOR with L_i to produce new R_i
- Set: L_i = R_{i-1}

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R_i = result of XOR

Step6: 32bits swap

Step7: Inverse Initial Permutation



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AES Overview

Unlike Feistel cipher, AES encrypts all 128 bits in one iteration. AES consists of three different types of layers, each of them manipulates all 128 bits of the data path (also called states).

(Each round, with the exception of the first, consists of all three layers.)

Key Addition layer

A 128-bit round key, or subkey, which has been derived from the main key in the key schedule, is XORed to the state (or data path).

Byte Substitution layer (S-Box)

Each element of the state is nonlinearly transformed (by providing confusion) using lookup tables with special mathematical properties.

Diffusion layer

It provides diffusion over all state bits. It further consists of two sublayers:

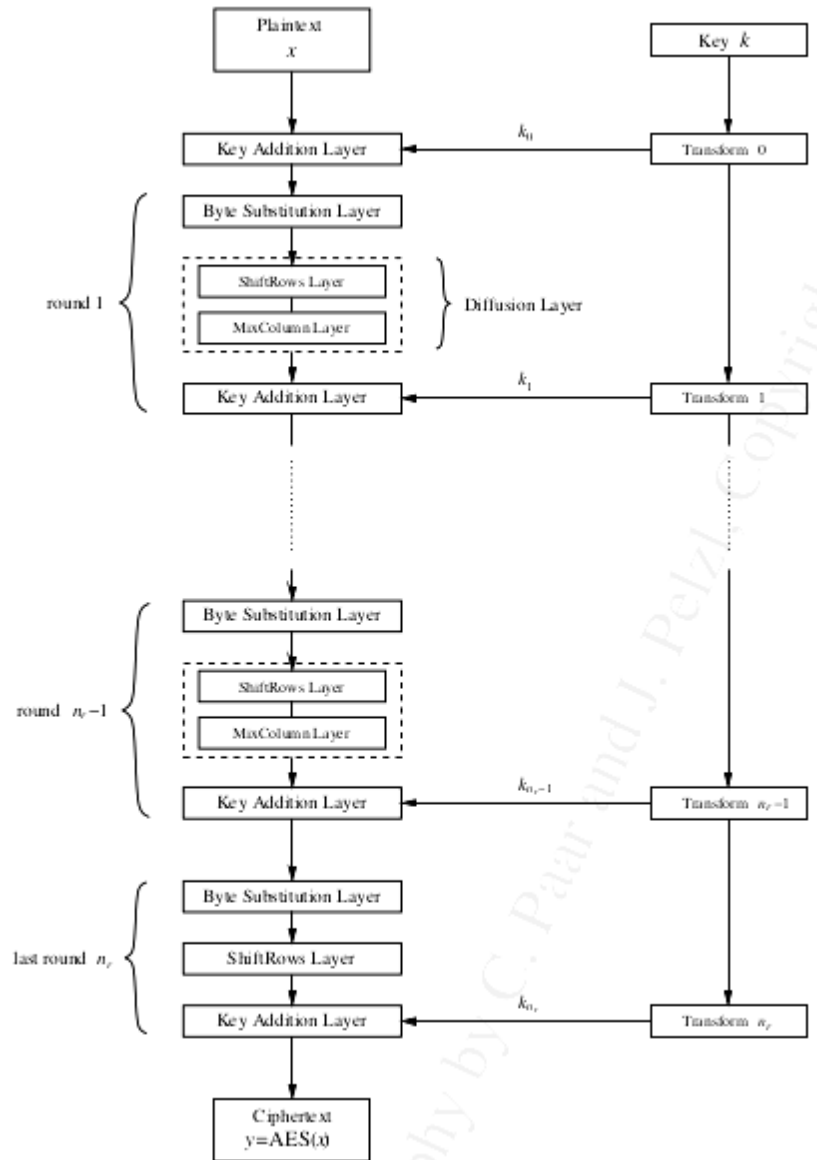
The **ShiftRows layer** permutes the data on a byte level.

The **MixColumn layer** combines blocks of four bytes.

Key Scheduling

Similar to DES, the key schedule computes round keys, or subkeys, $(k_0, k_1, \dots, k_{nr})$ from the original key.

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AES Internal Structure:

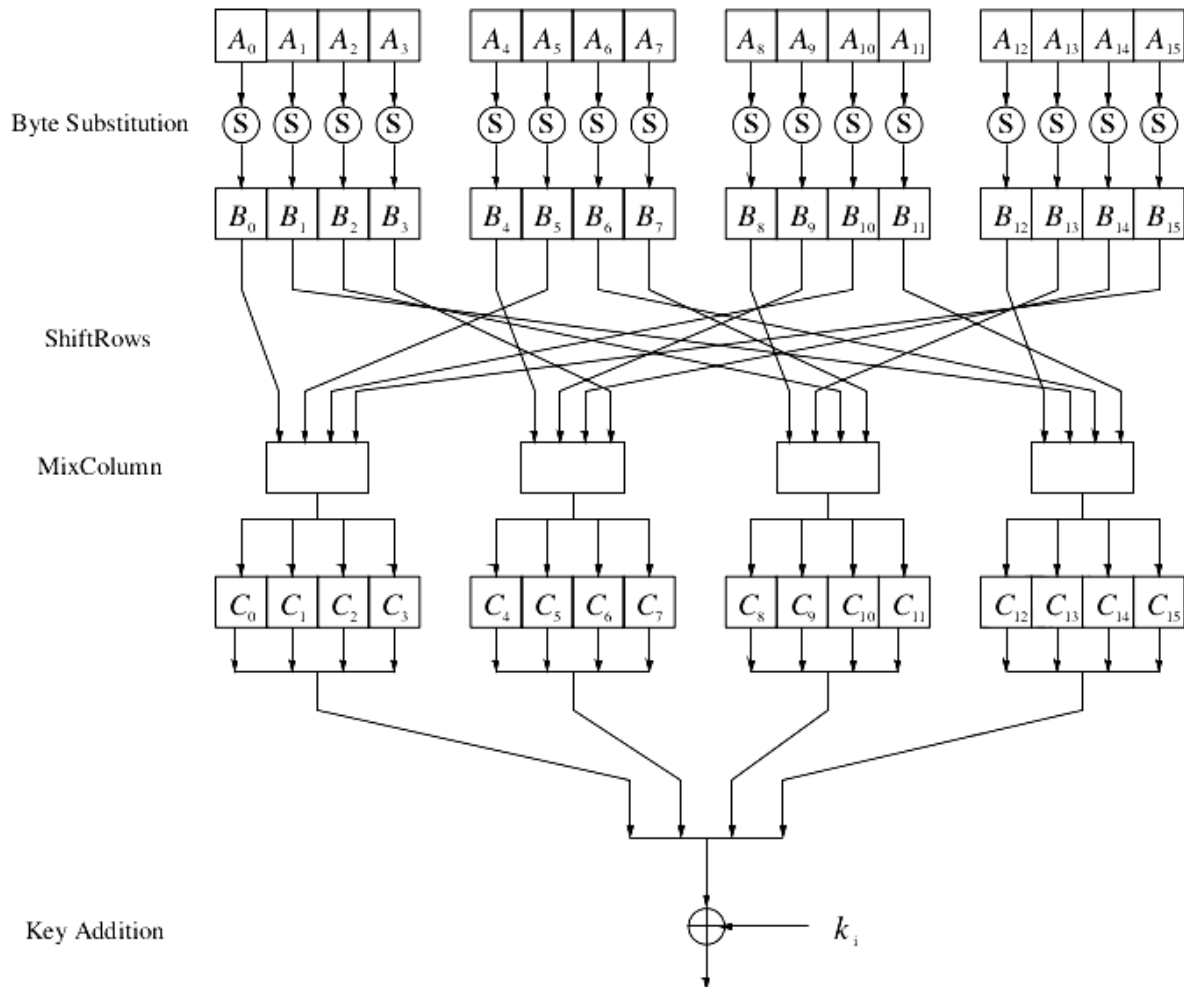
The 16-byte input $A_0, \dots, A_{15}$ is fed byte-wise into the S-Box.

The 16-byte output $B_0, \dots, B_{15}$ is permuted byte-wise in the ShiftRows layer and mixed by the MixColumn transformation $c(x)$.

Finally, the 128-bit subkey k_i is XORed with the intermediate result.

(Note : AES is a byte-oriented cipher.)

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Byte Substitution Layer

The Byte Substitution layer is viewed as a row of 16 parallel S-Boxes, each with 8 input and output bits.

(Note: all 16 S-Boxes are identical, unlike DES.)

Each state byte A_i is substituted by another byte B_i i.e. $S(A_i) = B_i$.

The S-Box substitution is a bijective mapping, i.e., each of the $2^8 = 256$ possible input elements is one-to-one mapped to one output element, by allowing to uniquely reverse the S-Box for decryption.

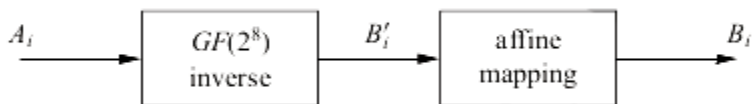
S-Box is usually realized as a 256-by-8 bit lookup table with fixed entries. e.g. $S((C2)_{\text{hex}}) = (25)_{\text{hex}}$
 $S(1100\ 0010) = (0010\ 0101)$

$GF(2^8)$ provides a high degree of nonlinearity and affine mapping “destroys” the algebraic structure of the Galois field.

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	y															
	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	63	7C	77	7B	F2	6B	6F	C5	30	01	67	2B	FE	D7	AB	76
1	CA	82	C9	7D	FA	59	47	F0	AD	D4	A2	AF	9C	A4	72	C0
2	B7	FD	93	26	36	3F	F7	CC	34	A5	E5	F1	71	D8	31	15
3	04	C7	23	C3	18	96	05	9A	07	12	80	E2	EB	27	B2	75
4	09	83	2C	1A	1B	6E	5A	A0	52	3B	D6	B3	29	E3	2F	84
5	53	D1	00	ED	20	FC	B1	5B	6A	CB	BE	39	4A	4C	58	CF
6	D0	EF	AA	FB	43	4D	33	85	45	F9	02	7F	50	3C	9F	A8
7	51	A3	40	8F	92	9D	38	F5	BC	B6	DA	21	10	FF	F3	D2
x 8	CD	0C	13	EC	5F	97	44	17	C4	A7	7E	3D	64	5D	19	73
9	60	81	4F	DC	22	2A	90	88	46	EE	B8	14	DE	5E	0B	DB
A	E0	32	3A	0A	49	06	24	5C	C2	D3	AC	62	91	95	E4	79
B	E7	C8	37	6D	8D	D5	4E	A9	6C	56	F4	EA	65	7A	AE	08
C	BA	78	25	2E	1C	A6	B4	C6	E8	DD	74	1F	4B	BD	8B	8A
D	70	3E	B5	66	48	03	F6	0E	61	35	57	B9	86	C1	1D	9E
E	E1	F8	98	11	69	D9	8E	94	9B	1E	87	E9	CE	55	28	DF
F	8C	A1	89	0D	BF	E6	42	68	41	99	2D	0F	B0	54	BB	16

{Mathematical description}



{Both steps for all 256 possible input elements of the S-Box are computed and stored to obtain the table.}

Asymmetric Key Cryptography:

Asymmetric key encryption (also called public-key cryptography) is a method of securing data using two mathematically related keys:

- Public Key → shared openly with everyone
- Private Key → kept secret by the owner

RSA (Rivest-Shamir-Adleman, 1977)

The first practical public-key encryption algorithm. It is widely used for encrypting small data (like session keys) and digital signatures.

Security relies on the difficulty of factoring large semiprimes (product of two primes). Modern keys are typically 2048-bit or 4096-bit.

RSA uses two exponents, **e** and **d**, where **e** is public and **d** is private. Suppose **P** is the plaintext and **C** is the ciphertext. Alice uses $C = P^e \text{ mod } n$ to create ciphertext **C** from plaintext **P**; Bob uses $P = C^d \text{ mod } n$ to retrieve the plaintext sent by Alice. The modulus **n**, a very large number, is created during the key generation process.

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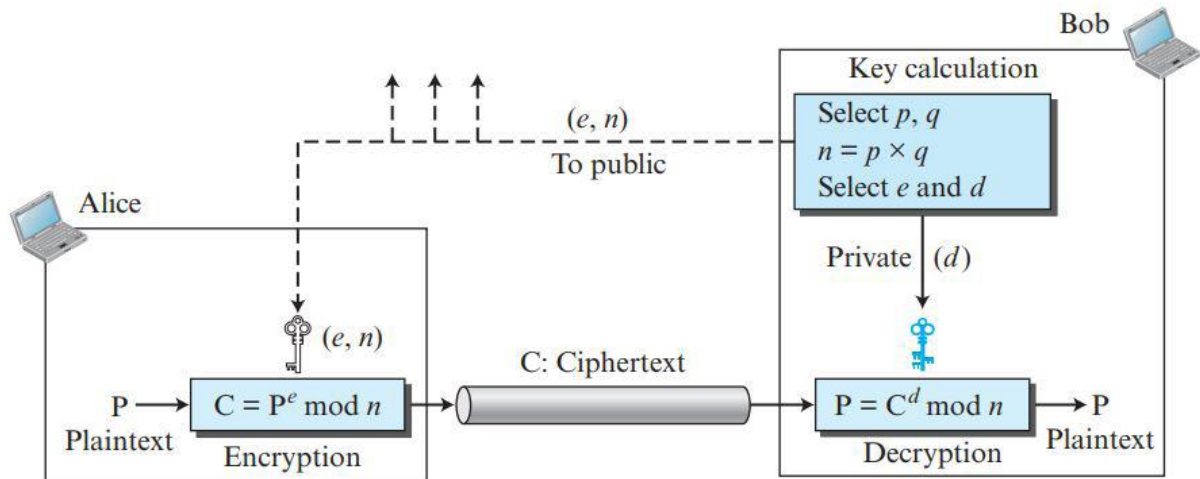


Fig. Encryption, decryption and key generation used in RSA

RSA: Key Generation:

The keys for RSA algorithm are generated in the following way:

1. Generate two large random prime numbers, p and q . (p and $q \geq 2^{512}$).
2. Compute $n = p \times q$, n is called modulus.
3. Compute the totient $\Phi(n) = (p-1)(q-1)$.
4. Choose e such that $1 < e < \Phi(n)$, where e and $\Phi(n)$ do not share factors other than one i.e. $\gcd(e, \Phi(n)) = 1$.
5. Compute d such that $d \times e = 1 \bmod \Phi(n)$ (i.e. $de \bmod \Phi(n) = 1$).
6. Publish e and n as the public key (Sender's public key) for encryption.
7. Keep d and n as private key (Receiver's private key) for decryption.

RSA Example:

- Choose two primes: $p = 61, q = 53$
- Compute $n = p \times q = 3233$ (this is part of both keys)
- Compute Euler's totient $\phi(n) = (p-1)(q-1) = 60 \times 52 = 3120$
- Choose public exponent $e = 17$ (coprime to $\phi(n)$)
- Compute private exponent d such that $(d \times e) \equiv 1 \bmod \phi(n) \rightarrow d$ is the modular inverse.
- Public key: $(e, n) = (17, 3233)$
- Private key: (d, n)

To encrypt a message m (as a number $< n$): ciphertext $c = m^e \bmod n$

To decrypt: $m = c^d \bmod n$

RSA: Key Distribution

- Suppose that Bob wants to send information to Alice. IF they decide to use RSA, Bob must know Alice's public key to encrypt the message and Alice must use her private key to decrypt the message.

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- To enable Bob to send his encrypted messages, Alice transmits her public key (e, n) to Bob via a reliable, but not necessarily secret, route. Alice's private key (d, n) is never distributed.

RSA: encryption and decryption

Given public key (e, n) and private key (d, n) are computed

- To encrypt bit pattern, P , compute cipher text C as:

$$C = P^e \bmod n$$

- To decrypt bit pattern, C , compute

$$P = C^d \bmod n$$

RSA Example:

Pick two prime numbers: $p=3$, $q=5$.

$$n = p \times q = 3 \times 5 = 15$$

$$\phi(n) = (p-1)(q-1) = (3-1)(5-1) = 2 \times 4 = 8.$$

Choose e satisfying $1 < e < \phi(n)$.

Let us choose $e=3$, which do not share any common factors with 8 rather than 1.

Compute d satisfying $de \bmod \phi(n)=1$

$$\text{So, } d \times 3 \bmod 8 = 1$$

Let us choose $d=11$ which satisfy the relation.

- So public key (e, n) is $(3,15)$ which is released publicly and the persons that want to send the message use this key to encrypt the message and send it to the receiver.

- Private key (d, n) is $(11,15)$ which is kept secret by the receiver.

Let us consider the message be 2.

So, at encryption process, the sender uses the public key to encrypt the message. Resulting cipher text will be:

$$C = m^e \bmod n = 2^3 \bmod 15 = 8.$$

At decryption process, the private key is used to decrypt the cipher text. Plain text is obtained as:

$$P = c^d \bmod n = 8^{11} \bmod 15 = 2.$$

Hence the original message 2 is obtained at receiver end after decryption.

Diffie-Hellman (DH) Key Exchange (1976)

This doesn't encrypt data directly but allows two parties to agree on a shared secret key over an insecure channel. It is a cornerstone of protocols like TLS.

- Both parties publicly agree on large prime p and base g .
- Each generates a private random number and computes a public value.
- They exchange public values and each computes the same shared secret using their private number and the other's public value.
- An eavesdropper cannot easily compute the shared secret due to the discrete log problem

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Steps for Diffie-Hellman Key Exchange

1. Public Parameter Agreement

Alice and Bob publicly agree on a large prime p and a generator g ($1 < g < p$).

Everyone (including Eve) knows p and g .

2. Private Key Selection

Alice secretly chooses a random private integer a (her private exponent), where $1 < a < p$.

Bob secretly chooses a random private integer b (his private exponent), where $1 < b < p$.

3. Public Key Computation and Exchange

Alice computes her public value: $A = g^a \bmod p$ and sends A to Bob.

Bob computes his public value: $B = g^b \bmod p$ and sends B to Alice.

(Eve sees A and B , along with p and g .)

4. Shared Secret Computation

Alice receives B and computes the shared secret: $K = B^a \bmod p$

Bob receives A and computes the shared secret: $K = A^b \bmod p$

Why does this work?

Because of the mathematical property of exponents:

$$(g^b)^a \bmod p = g^{(b*a)} \bmod p$$

and

$$(g^a)^b \bmod p = g^{(a*b)} \bmod p$$

Since multiplication is commutative ($ab = ba$), both sides produce the exact same value $K = g^{(a*b)} \bmod p$.

Alice and Bob now share the same secret K , which can be used (often after hashing) as a symmetric session key.

Eve knows p , g , A , and B , but to find K she would need to solve the discrete logarithm problem—finding a from $A = g^a \bmod p$ (or b from B). For sufficiently large p (e.g., 2048 bits or more), this is computationally infeasible with classical computers.

Example:

Here's a well-known small example to see it in action (in real systems, p is huge—thousands of bits—so don't use small numbers for security!):

- Public parameters: $p = 23$ (prime), $g = 5$ (primitive root modulo 23)
- Alice chooses private key $a = 4$
She computes $A = 5^4 \bmod 23 = 625 \bmod 23 = 4$
Sends $A = 4$ to Bob.

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Bob chooses private key $b = 3$

He computes $B = 5^3 \bmod 23 = 125 \bmod 23 = 10$

- Sends $B = 10$ to Alice.
- Alice computes shared secret: $K = B^a \bmod 23 = 10^4 \bmod 23 = 10000 \bmod 23 = 18$
- Bob computes shared secret: $K = A^b \bmod 23 = 4^3 \bmod 23 = 64 \bmod 23 = 18$

Both arrive at $K = 18$. They can now use this (or a derived key) for symmetric encryption.

Elliptic Curve:

An elliptic curve over a finite field is defined by this equation:

$$y^2 = x^3 + ax + b \pmod{p}$$

where:

- x and y are coordinates of points on the curve,
- a and b are constants that define the specific curve,
- p is a large prime number (the modulus),
- All operations are performed modulo p .

When plotted (over real numbers for visualization), it looks like a smooth, symmetric curve. In cryptography, we work with a **discrete set of points** satisfying the equation over the finite field. There is also a special **point at infinity** (denoted O), which acts as the identity element (like zero in addition).

Digital Signatures and Certificates:

Digital signatures and digital certificates are two fundamental technologies used to ensure **security, authenticity, and trust** in online communication. They are widely used in areas such as online banking, secure emails, software distribution, e-commerce, and electronic document signing.

By providing **authentication, integrity, confidentiality, and non-repudiation**, these technologies help protect data from tampering, impersonation, fraud, and unauthorized access in digital environments.

Digital Signature

A **digital signature** is a cryptographic technique used to verify the **authenticity, integrity, and non-repudiation** of a digital message or document. It ensures that the message was created by a known sender and that it has not been altered during transmission.

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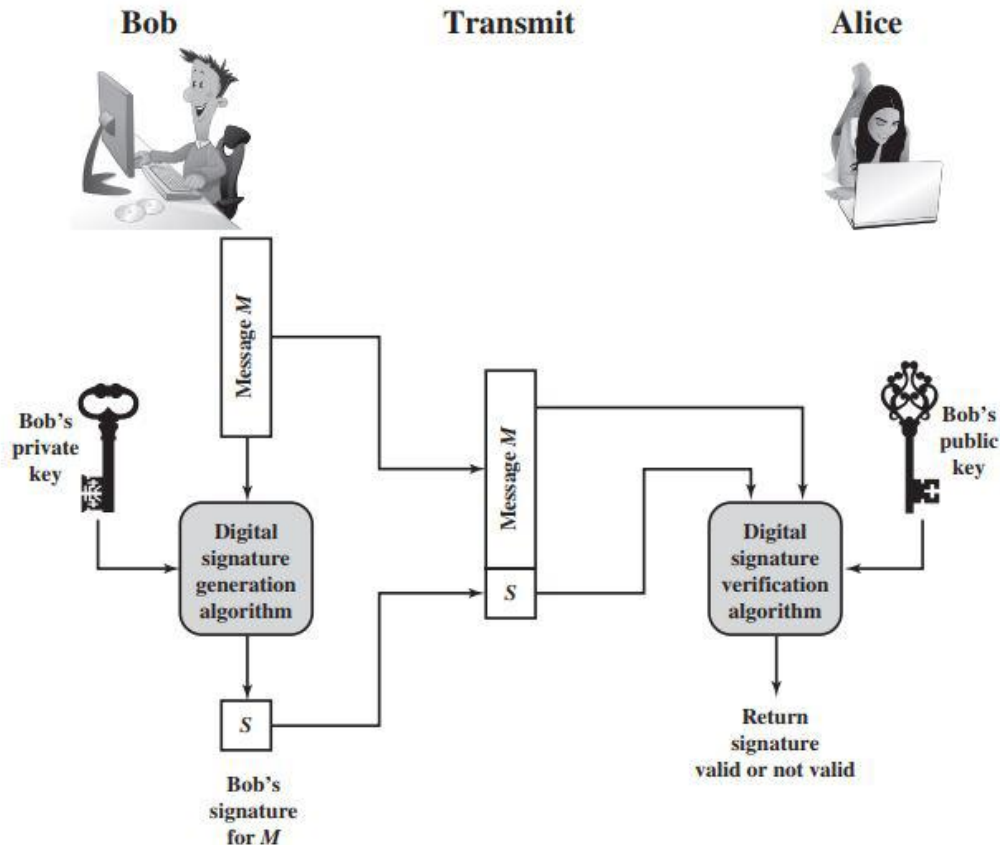


Figure: Generic Model of digital signature

Components of Digital Signature:

1. Key Generation Algorithm

Digital signatures use **asymmetric cryptography**, which involves a pair of keys:

- **Private Key:** Kept secret by the owner and used to create the signature.
- **Public Key:** Shared with others and used to verify the signature.

This key pair ensures secure authentication during digital transactions.

2. Signing Algorithm

To create a digital signature:

1. A **hash function** is applied to the original message to generate a fixed-length value called a **message digest**.
2. This message digest is then **encrypted using the sender's private key**.
3. The encrypted hash value forms the **digital signature**.

Instead of encrypting the entire message, only the hash is encrypted because:

- Hash values are much shorter
- Hashing is faster than encryption
- It improves efficiency without reducing security

3. Signature Verification Algorithm

At the receiver's side:

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1. The digital signature is **decrypted using the sender's public key**, producing the original message digest.
2. The receiver independently computes the hash of the received message using the same hash function.
3. Both hash values are compared:
 - If they match → signature is **valid**
 - If they differ → message integrity is compromised

Working Principle:

The process of creating and verifying a digital signature involves the following steps:

1. The sender computes a **message digest** using a one-way hash function.
2. The message digest is encrypted using the **sender's private key**, forming the digital signature.

Digital Signature = Encryption (Sender's Private Key, Message Digest)

3. The sender transmits:
 - Original message
 - Digital signature
4. The receiver decrypts the digital signature using the **sender's public key**.
5. The receiver computes a fresh message digest from the received message.
6. If both message digests are identical, **integrity and authenticity are verified**.

A one-way hash function ensures that:

- Hash computation is easy
- Retrieving the original message from the hash is computationally infeasible

Public Key Infrastructure:

Public Key Infrastructure (PKI) is a security framework that issues and manages digital certificates to establish trust on digital networks. It ensures secure communication by validating identities, protecting data, and preventing unauthorized access.

- Provides unique digital identities to users, devices, and systems
- Secures communication using public and private key pairs
- Prevents MITM attacks by verifying key ownership
- Ensures confidentiality, integrity, and authenticity of data

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nu3.2 Network Security

Transport Layer Security: SSL

- SSL stands for Secure Socket Layer.
- SSL originally developed by NetScape in 1994.
- SSL v3, designed with public input, became Internet standard known as Transport Layer Security (TLS).
- SSL is a general-purpose service implemented as a set of protocols that rely on TCP.
- It provides security at transport layer.
- SSL/TLS provides confidentiality using symmetric encryption, data integrity and message authentication between server and client.

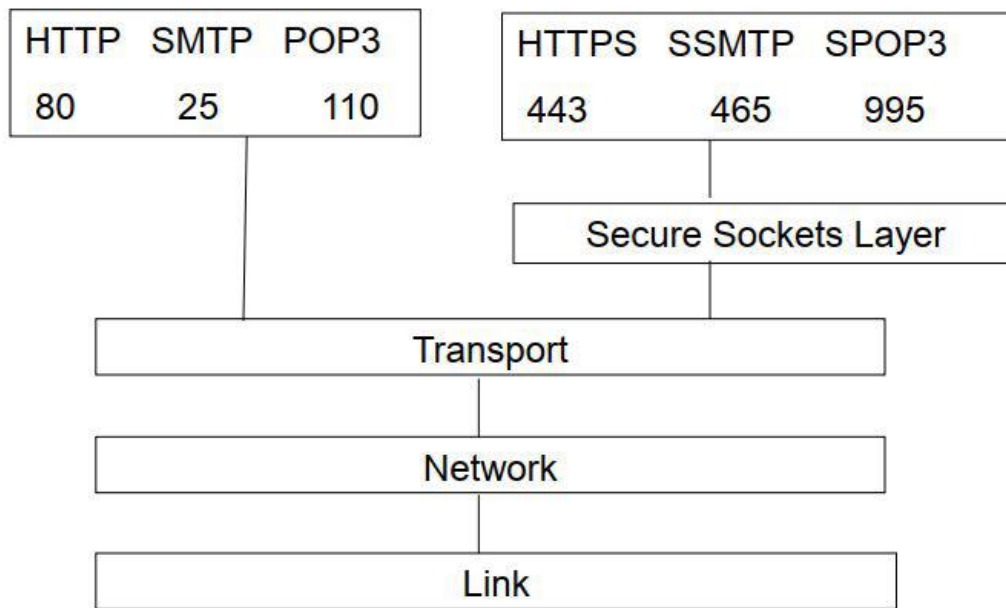


Figure: SSL uses.

SSL Architecture

- SSL has two layers of protocols which are implemented above the TCP layer.
- The SSL Record Protocol provides basic security services to various higher layer protocols e.g. HTTP.
- Three higher-layer protocols are defined as part of SSL:
 - i. Handshake protocol
 - ii. Change cipher Spec Protocol
 - iii. Alert Protocol

Two important concepts of SSL:

i. SSL connection:

- A transient peer-to-peer communication link that provides a suitable type of service.
- Every connection is associated with one session.

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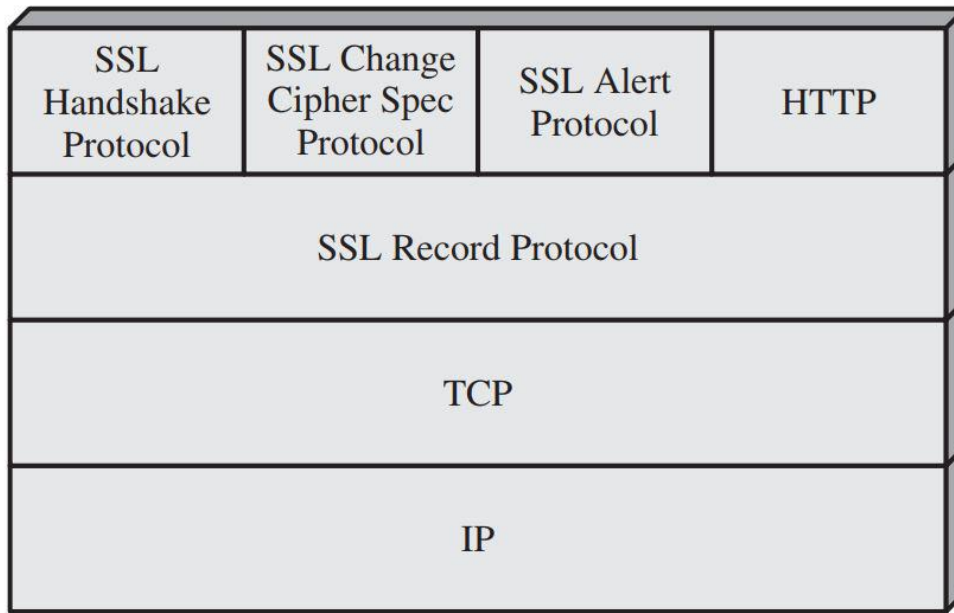


Figure: SSL protocol stack

ii. SSL session:

- An SSL session is an association between a client and a server.
- Created by handshake protocol.

SSL Record Protocol

- SSL Record Protocol provides two services for SSL connections:
 - Confidentiality:** The Handshake Protocol defines a shared secret key that is used for conventional encryption of SSL payloads.
 - Message Integrity:** The Handshake Protocol also defines a shared secret key that is used to form a message authentication code (MAC).

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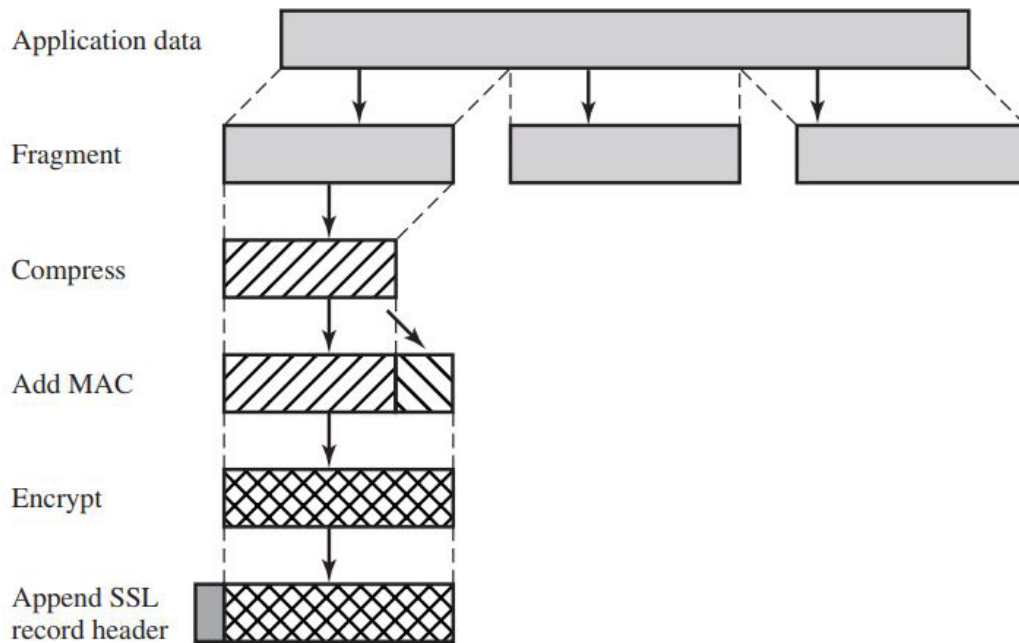


Figure: SSL protocol operation

SSH Handshake Protocol:

- Allows server and client to:
- Authenticate each other
- To negotiate encryption and MAC (Message Authentication Code) algorithms
- To negotiate cryptographic keys to be used

Comprises a series of messages exchanged in phases

- Establish security capabilities
- Server authentication and key exchange
- Client authentication and key exchange
- finish

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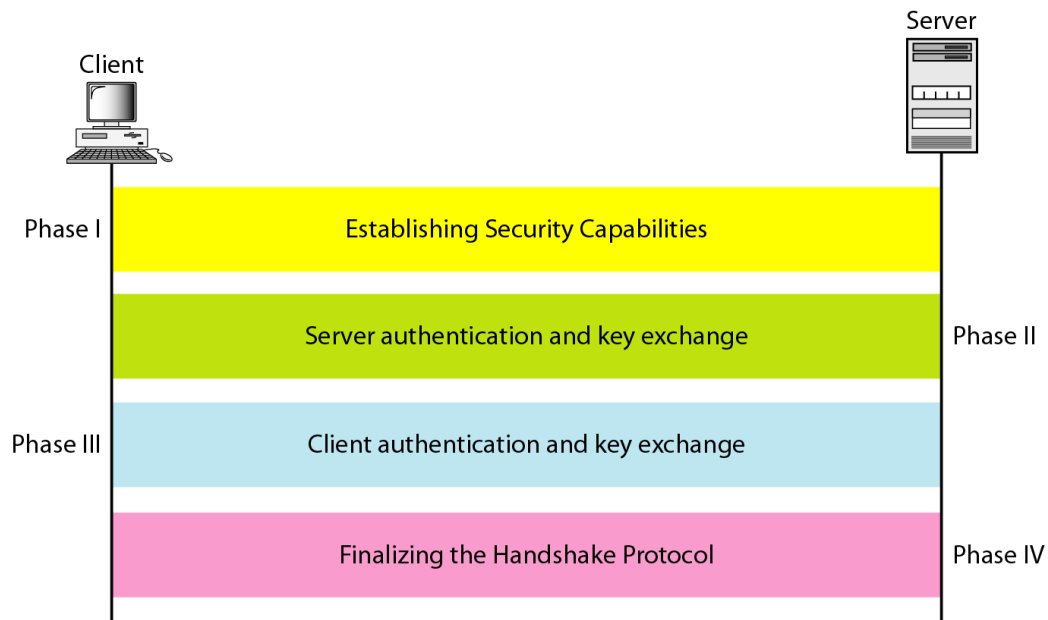


Figure: SSL handshake protocol

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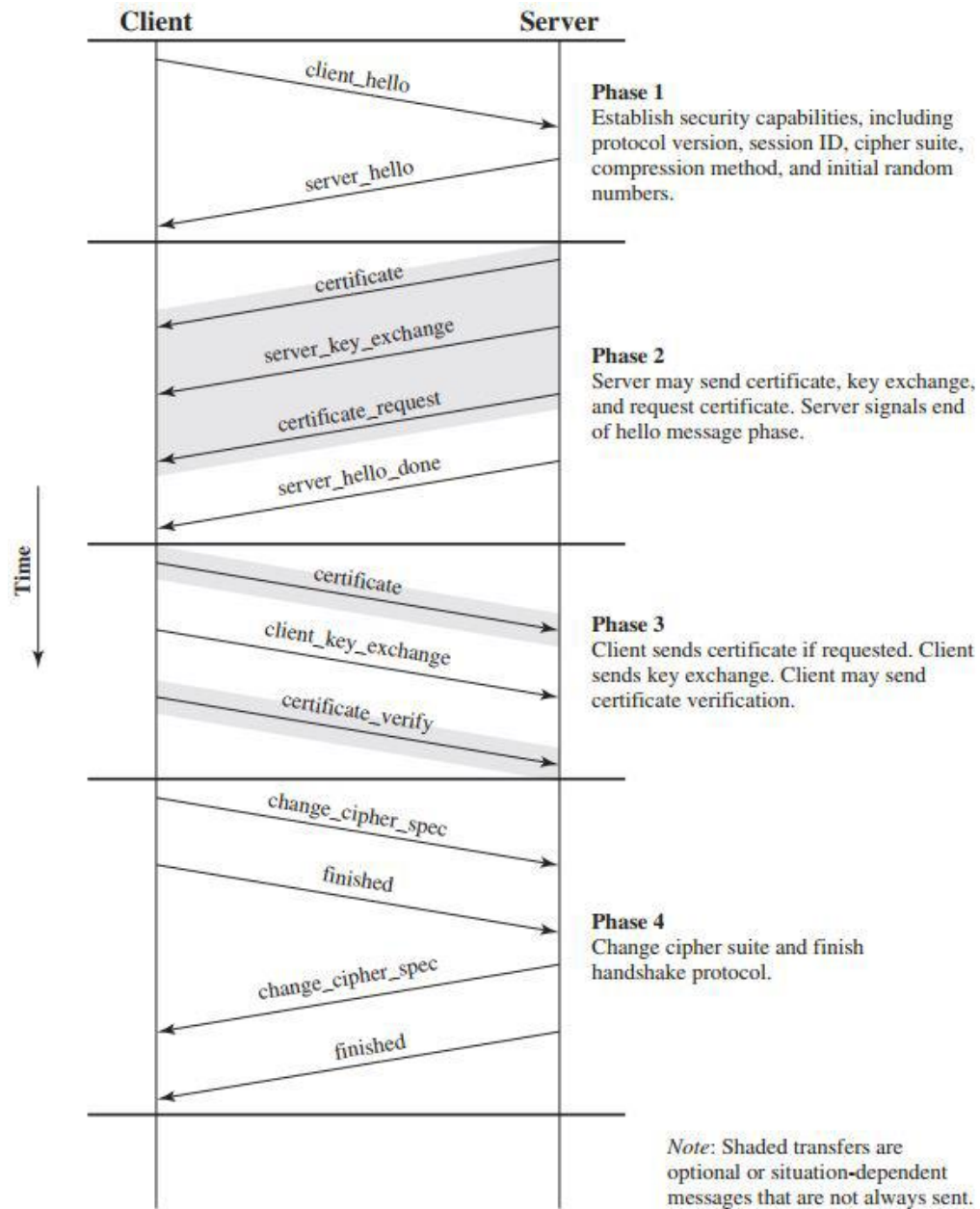


Figure: SSL Handshake protocol action

SSL Change Cipher Spec Protocol:

- One of 3 SSL specific protocols which use the SSL Record protocol.
- A single message with the value of 1.
- Causes pending state to become current, hence updating the cipher suit in use.

SSL Alert Protocol:

- Used to convey SSL-related alerts to the peer entity.
- SSL alerts are compressed and encrypted like all SSL data.

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- Each message in this protocol consists of two bytes. The first byte takes the value warning (1) and fatal (2) to convey the severity of the message. If the level is fatal, SSL immediately terminates the connection.
- **Specific alert:** Unexpected message, decompression failure, handshake failure, illegal parameters.

Transport Layer Security (TLS)

Transport Layer Security (TLS) is a cryptographic protocol designed to provide security at the transport layer. It was derived from a security protocol called Secure Socket Layer (SSL). TLS ensures that no third party can eavesdrop on or tamper with any message transmitted between a client and a server.

Benefits of TLS

1. **Encryption:** TLS/SSL helps secure transmitted data using encryption, ensuring that sensitive information remains confidential and protected from unauthorized access.
2. **Interoperability:** TLS/SSL works seamlessly with most web browsers, including Microsoft Internet Explorer, and operates across a wide range of operating systems and web servers.
3. **Algorithm Flexibility:** TLS/SSL provides flexibility in choosing authentication mechanisms, encryption algorithms, and hashing algorithms used during secure sessions, allowing systems to adapt to different security requirements.
4. **Ease of Deployment:** Many applications can temporarily implement TLS/SSL on Windows Server 2003 operating systems, making it relatively easy to deploy in various environments.
5. **Ease of Use:** Since TLS/SSL operates beneath the application layer, most of its processes are completely invisible to the client, ensuring a smooth and user-friendly experience.

Working of TLS

When a client connects to a server using TCP, it initiates a process called the SSL/TLS handshake. During this handshake, the client sends several pieces of information to the server, including:

- The version of SSL/TLS it supports.
- A list of cipher suites (encryption algorithms) it can use.
- The compression methods it supports.

This allows the server to select compatible settings and establish a secure connection.

The server checks the highest SSL/TLS version supported by both parties, selects a cipher suite from the client's options (if compatible), and optionally chooses a compression method. After this setup, the server provides its digital certificate, which must be trusted either directly by the client or by a trusted third party.

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Once the certificate is verified and the client is assured that the server is legitimate (preventing a man-in-the-middle attack), a key exchange occurs. This can involve a public key, a Premaster Secret, or no key exchange at all, depending on the chosen cipher suite.

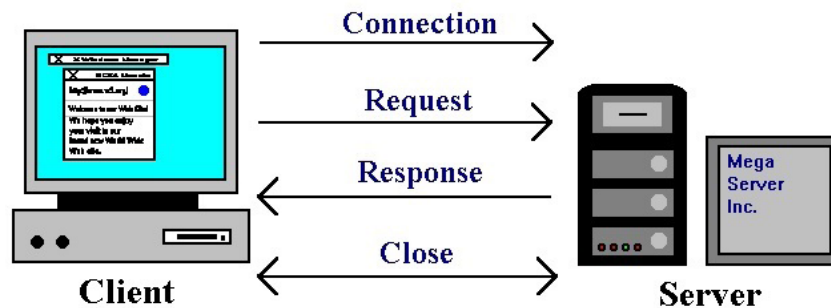
Both the server and client then compute the session key for symmetric encryption. Once the handshake is complete, the two hosts can communicate securely. If the TCP connection is closed improperly, both sides will know the connection was interrupted—but it cannot be compromised, only terminated.

Hyper Text Transfer Protocol (HTTP):

Being a stateless application-layer protocol, HTTP does not retain session information between requests, which limits its ability to handle complex client-server interactions without additional mechanisms like cookies or sessions.

- Hypertext Transfer Protocol (HTTP) is a protocol used which transfer hypertext over the Web.
- HTTP has been the most widely used protocol for data transfer over the Web.
- Hyper-text exchanged using HTTP goes as plain text which makes it less secure.
- The web server delivers data to the user in the form of web pages when the user initiates an HTTP request.
- Connection between the web browser and the server ends after the transaction is finished. This makes HTTP a stateless protocol.

Hypertext Transport Protocol (HTTP)



Stages in an HTTP Transaction

Gregory S. Aist
April 28, 1995

Figure: HTTP transaction

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Advantages of HTTP:

- Because fewer connections are running at once, it delivers reduced CPU and memory utilization.
- It allows requests and answers to be pipelined via HTTP.
- Because there are fewer TCP connections, it provides less network congestion.
- During the first stage of connection establishment, handshakes are exchanged. Because there is no handshaking, it provides lower latency for subsequent requests.
- Without terminating the TCP connection, it reports problems.

Disadvantages of HTTP:

- It is applicable to point-to-point connections.
- It isn't mobile-friendly.
- It sends more data than needed.
- It doesn't provide trustworthy exchange (in the absence of retry mechanism).
- When the client receives all the data it requires, the connection is not terminated. Therefore, the server won't be accessible during this time

Hypertext Transfer Protocol Secure (HTTPS)

HTTPS ensures end-to-end encryption and authentication by leveraging TLS/SSL, safe guarding data from eavesdropping and tampering during transmission.

- HTTPS is used for secure communication as Hypertext Transfer Protocol (HTTP) doesn't.
- Data can be transferred using this protocol in an encrypted format.
- The HTTPS protocol is mostly utilized in situations when entering login credentials is necessary. Modern browsers like Chrome distinguish between the HTTP and HTTPS protocols based on distinct markings.
- HTTPS employs an encryption mechanism called Secure Sockets Layer (SSL), also known as Transport Layer Security, to enable encryption.

Advantages of HTTPS

- Provides in-transit data security.
- Shields your website from data breaches, phishing, and MITM attacks.
- Increases the visitors' trust to your website.
- Eliminates the "NOT Secure" alerts.
- Assist you in raising your website's ranking.

Disadvantages of HTTPS

- While switching to HTTPS SSL certificate needs to be bought.
- Encrypting and decrypting data across HTTPS connections requires a lot of computation.
- There will be issues with caching some information over HTTPS. Public caching of those that previously took place won't happen again.
- Certain proxy servers and firewalls prevent users from accessing HTTPS websites. Both deliberate and inadvertent actions might result from this.

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- If configuration issues encounter HTTP will be used by your website to obtain files rather than HTTPS.

HTTP vs HTTPS

Feature	HTTP	HTTPS
Security	Data is sent as plain text — easy to intercept.	Data is encrypted — safer from hackers.
Port	Uses port 80 by default.	Uses port 443 by default.
Trust	Browsers may label it “Not Secure.”	Shows a padlock in the address bar.
Speed	Slightly slower on old systems.	Often faster due to newer protocols and optimizations.
Use case	Suitable for public or non-sensitive info.	Best for sensitive data like payments, logins, personal info.

Why Choose HTTPS over HTTP?

1. Security

HTTPS is like sending your message in a locked box that only the receiver can open. This keeps sensitive information, like passwords or credit card numbers, safe from hackers.

2. Trust & Authority

Web browsers show a padlock icon for HTTPS websites. People trust these sites more, and search engines also rank them higher. So, HTTPS can help you win both visitors’ trust and better search results.

3. Speed & Better Tracking

HTTPS websites load faster than HTTP also helps website owners see where their visitors come from (like social media or ads) more accurately, which is useful for improving their content.

IPSec

- IPSec stands for Internet Protocol Security.
- It is a collection of protocols to provide security for a packet at the network layer.
- IPSec is a standard framework for ensuring private communication over public network.
- It has become the most common network layer security control, typically used to create a VPN.

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- A VPN is a virtual network built on top of existing physical networks that can provide a secure communication for data transfer.
- VPNs are used most often to protect communication carried over public networks such as the Internet.
- A VPN can provide several types of data protection:
 - _ Confidentiality
 - _ Integrity
 - _ Data origin authentication

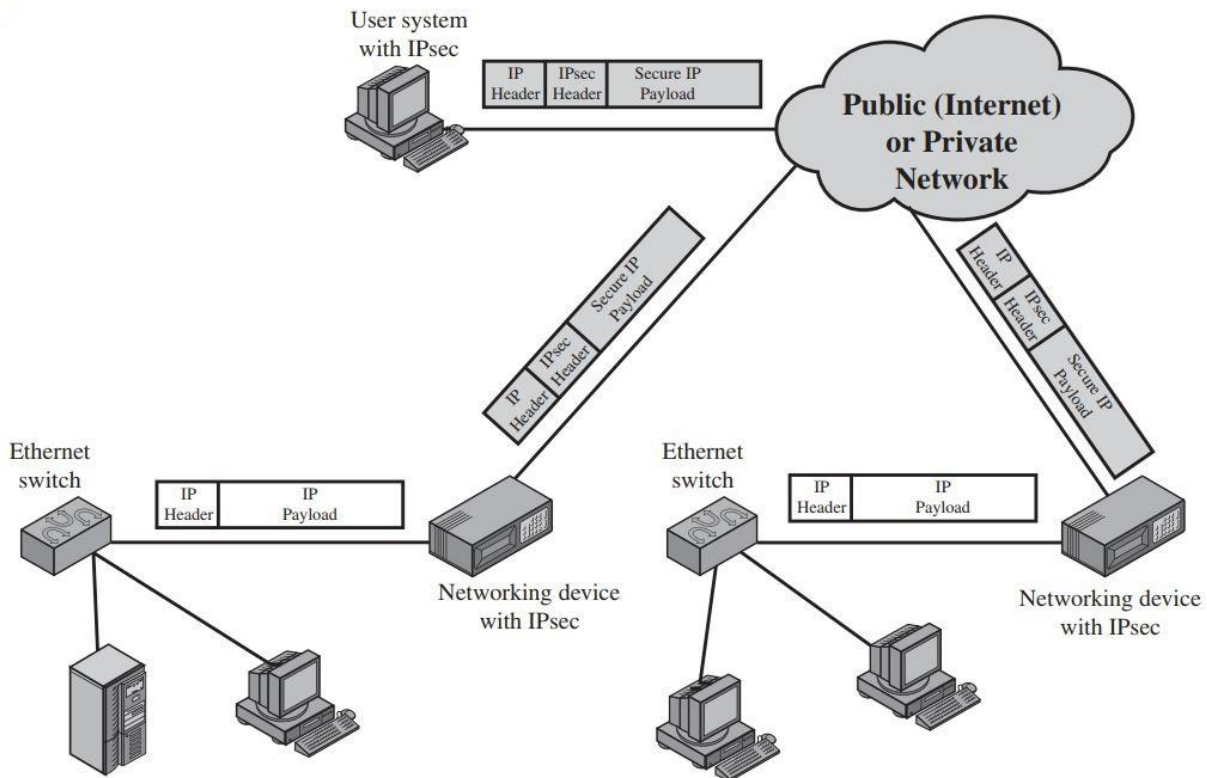


Figure: An IPSec scenario.

Modes of IPSec

IPSec operates in one of two different mode: transport mode and tunnel mode. Transport Mode

- Transport mode protects the payload to be encapsulated in the network layer i.e. it protects what is delivered from the transport layer to the network layer.

- It does not protect the whole IP header or in other words, it does not protect the whole IP packet.

Tunnel Model

- In tunnel mode, IPSec protects the entire IP packet. It takes an IP packet, including the header, applies IPSec security methods to the entire packet, and then adds a new IP header

8.16.1.2 IPSec Components

- Two security protocols :

- Authentication Header (AH): AH is an extension header to provide message authentication. Because message authentication is provided by ESP, the use of AH is deprecated.

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- Encapsulating Security Payload (ESP): ESP consists of an encapsulating header and trailer used to provide encryption or combined encryption/authentication and integrity protection
- Internet Key Exchange (IKE) protocol
- IPsec uses IKE to negotiate IPsec connection settings.
- Authenticate endpoints to each other.
- Define the security parameters of IPsec-protected connections
- Negotiate secret keys and

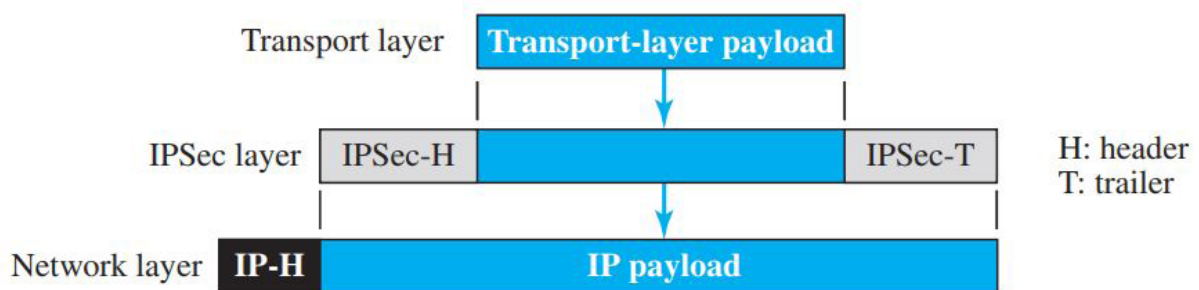


Figure: An IPsec in Transport mode

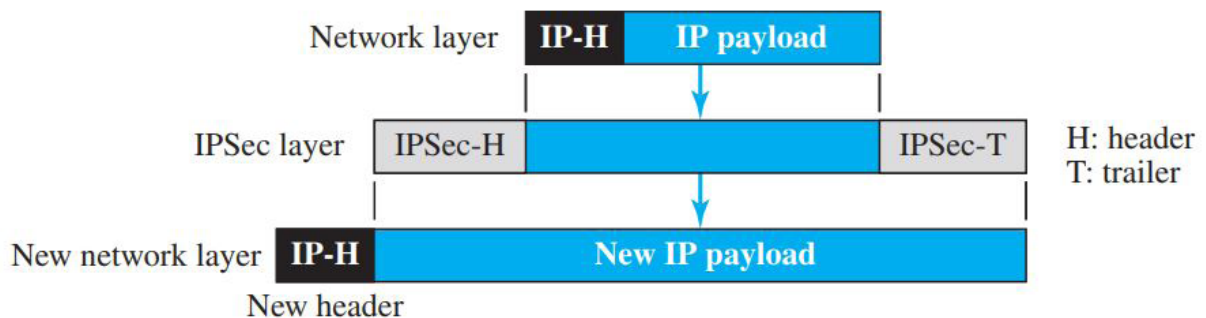


Figure: An IPsec in Tunnel mode

Manage, update, and delete IPsec-protected communication channels.

- IP Payload Compression Protocol (IPComp)
- Optionally, Ipsec can use Ipcomp to compress packet payloads before encrypting them.

Benefits of IPsec

- When IPsec is implemented in a firewall or router, it provides strong security that can be applied to all traffic crossing the perimeter. Traffic within a company or workgroup does not incur the overhead of security-related processing.
- IPsec in a firewall is resistant to bypass if all traffic from the outside must use IP and the firewall is the only means of entrance from the Internet into the organization.

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- IPsec is below the transport layer (TCP, UDP) and so is transparent to applications. There is no need to change software on a user or server system when IPsec is implemented in the firewall or router. Even if IPsec is implemented in end systems, upper-layer software, including applications, is not affected.
- IPsec can be transparent to end users. There is no need to train users on security mechanisms, issue keying material on a per-user basis, or revoke keying material when users leave the organization.
- IPsec can provide security for individual users if needed

IPSec Applications

- Secure branch office connectivity over the Internet.
- Secure remote access over the internet.
- Establishing extranet and intranet connectivity with partners.

Firewall

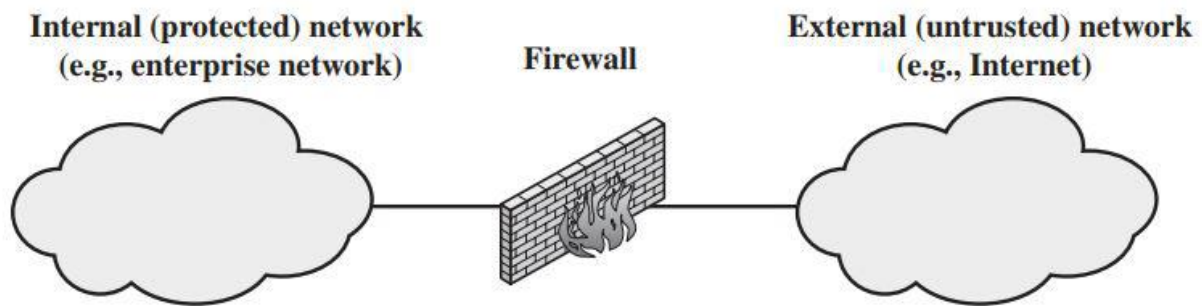


Figure 8.31: A general model of firewall.

- Firewall is a device (usually a router or computers) installed between the internal network of an organization and the rest of the Internet.
- A firewall is a network security system designed to prevent unauthorized access to or from a private network.
- A firewall security policy dictates which traffic is authorized to pass in each direction.
- A firewall may be designed to operate as a filter at the level of IP packets, or may operate at a higher protocol layer.
- A firewall provides a location for monitoring security-related events as a single-choke point.
- Auditing and controlling access can implement alarms for abnormal behavior.

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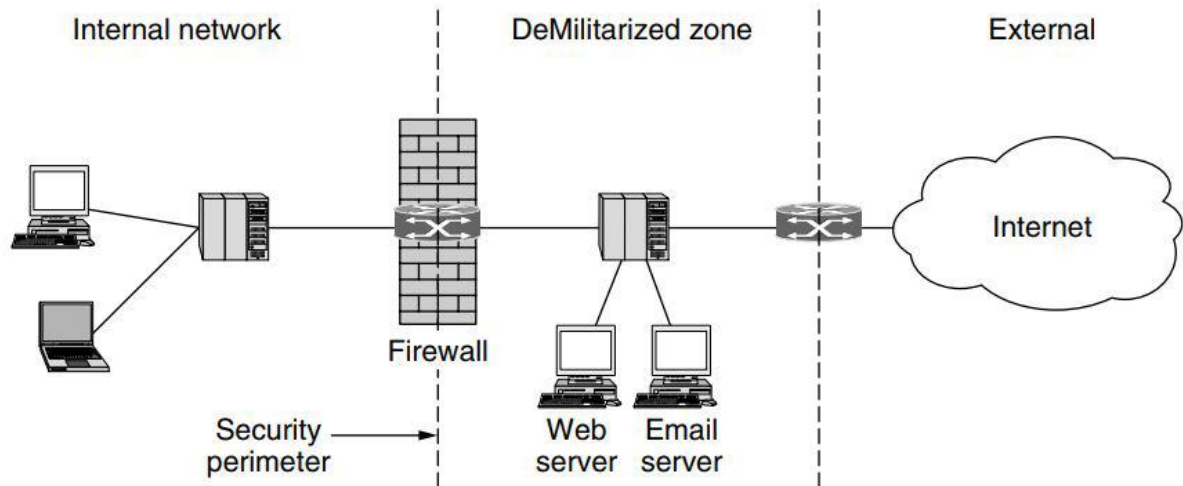


Figure 8.32: A firewall protecting an internal network.

Firewall Design Goals

- All traffic from inside to outside, and vice-versa, must pass through the firewall. This is achieved by physical blocking all access to the local network except via the firewall.
- Only authorized traffic, as defined by the local security policy, will be allowed to pass. Various types of firewalls are used, which implement various types of security policies.
- The firewall itself is immune to penetration; trusted system with a secured OS used to host a firewall.

Firewall Control Access Methods

Service Control

- Filter traffic on the basis of IP address, protocol or TCP port address.
- Example: block port 80, allow port 23.

Directional Control

- Determines the direction in which particular service requests may be initiated and allowed to flow through the firewall.

User control

- Controls access to a service according to which user is attempting to access it.
- This feature is typically applied to users inside the firewall perimeter (local users). It may also be applied to incoming traffic from external users.

Behavior control

- Controls how particular services are used. For example, the firewall may filter e-mail to eliminate spam, or it may enable external access to only.

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Scope of Firewall

The following capabilities are within the scope of a firewall:

- i. A firewall defines a single choke point that keeps unauthorized users out of the protected network, prohibits potentially vulnerable services from entering or leaving the network, and provides protection from various kinds of IP spoofing and routing attacks. The use of a single choke point simplifies security management because security capabilities are consolidated on a single system or set of systems.
- ii. A firewall provides a location for monitoring security-related events. Audits and alarms can be implemented on the firewall system.
- iii. A firewall is a convenient platform for several Internet functions that are not security related. These include a network address translator, which maps local addresses to Internet addresses, and a network management function that audits or logs Internet usage.
- iv. A firewall can serve as the platform for IPsec. Using the tunnel mode capability the firewall can be used to implement virtual private.

Types of Firewalls

- Packet-Filter Firewall
- Proxy Firewall

1 Packet Filter Firewall

- Packet-Filter Firewall applies a set of rules to each incoming and outgoing IP packet and then forwards or discards the packet.
- Packet filter is based on
 - source IP address,
 - destination IP address,
 - source port address,
 - destination port address,
 - IP protocol,
 - Interface
- The packet filter is typically set up as a list of rules based on matches to fields in the IP or TCP.

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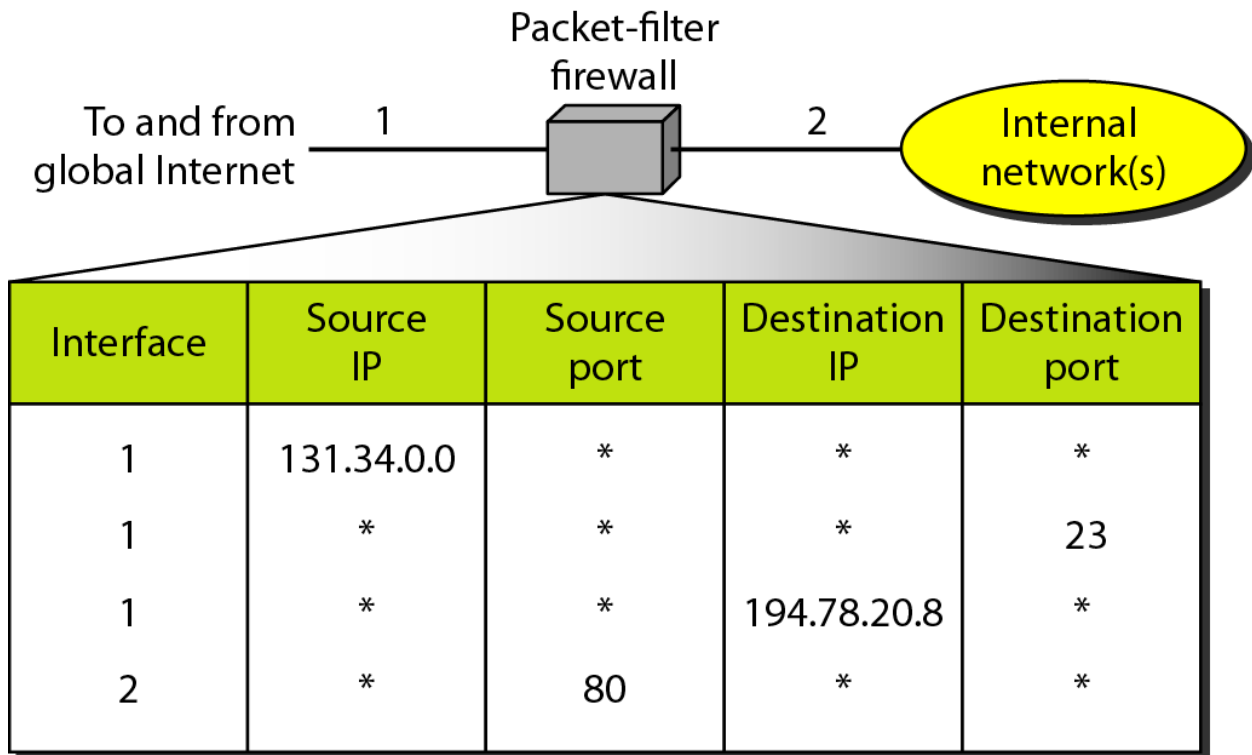


Figure: An example of packet filter firewall.

According to the figure, the following packets are filtered:

- i. Incoming packets from network 131.34.0.0 are blocked (security precaution). Note that the _____ (asterisk) means “any.”
- ii. Incoming packets destined for any internal TELNET server (port 23) are blocked.
- iii. Incoming packets destined for internal host 194.78.20.8 are blocked. The organization wants this host for internal use only.
- iv. Outgoing packets destined for an HTTP server (port 80) are blocked. The organization does not want employees to browse the Internet.

Application Gateway Firewall

- Also called application proxy firewall or simply proxy firewall.
- Acts as a relay of application-level traffic.
- It inspects the message of application.

Example: As an example, assume that an organization wants to implement the following policies regarding its web pages:

only those Internet users who have previously established business relations with the company can have access to other users must be blocked. In this case, a packet-filter firewall is not feasible because it cannot distinguish between different packets arriving at TCP port 80 (HTTP). Testing must be done at the application level (using URLs).

The user contacts the gateway using a TCP/IP application, such as HTTP or FTP, and the gateway asks the user for the name of the remote host to be accessed. When the user responds and provides

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a valid user ID and authentication information, the gateway contacts the application on the remote host and relays TCP segments containing the application data between the two endpoints.

Limitation of Firewalls

- **IP Snooping:** Routers can't know if data "really" comes from claimed source. An intruder can transmit packets from the outside with source IP address field containing an address of an internal host.
- The firewall does not protect against internal threats, Such as a dishonest employee or an employee who unwittingly cooperates with an external attacker.
- The firewall cannot protect against the transfer of virus-infected programs or files.

VPN: Virtual Private Network

A virtual Private Network (VPN) is a security technology that creates an encrypted tunnel between your device and a VPN server over the internet, so your traffic travels privately and your real IP address is hidden.

- **Privacy protection:** Hides your IP and encrypts traffic, so ISPs, advertisers, and third parties can't easily monitor your browsing.
 - **Security on public Wi-Fi:** Encryption protects logins and personal data on unsafe networks (airports, cafes).
 - **Bypass geo-restrictions:** Changes your apparent location by routing through another region's server, unlocking region-limited sites or services.
 - **Reduce ISP throttling:** Since traffic is encrypted, ISPs can't easily identify specific activities (streaming/gaming) to selectively slow them.
 - **Secure remote access:** Let employees connect to internal company resources securely from outside the organization.
-
- One of the applications of IPSec is in virtual private networks.
 - VPN creates a network that is private but virtual. It is private because it guarantees privacy inside the organization. It is virtual because it does not use real private WANs; the network is physically public but virtually private.
 - It is built on top of existing physical networks and provides a secure communications mechanism for data and IP information transmitted between networks.
 - VPN technology uses the ESP protocol of IPSec in the tunnel mode. Moreover, firewalls, - - VPNs, and IPsec with ESP in tunnel mode are a natural combination and widely used in practice.
 - VPSs can use both symmetric and asymmetric forms of cryptography.

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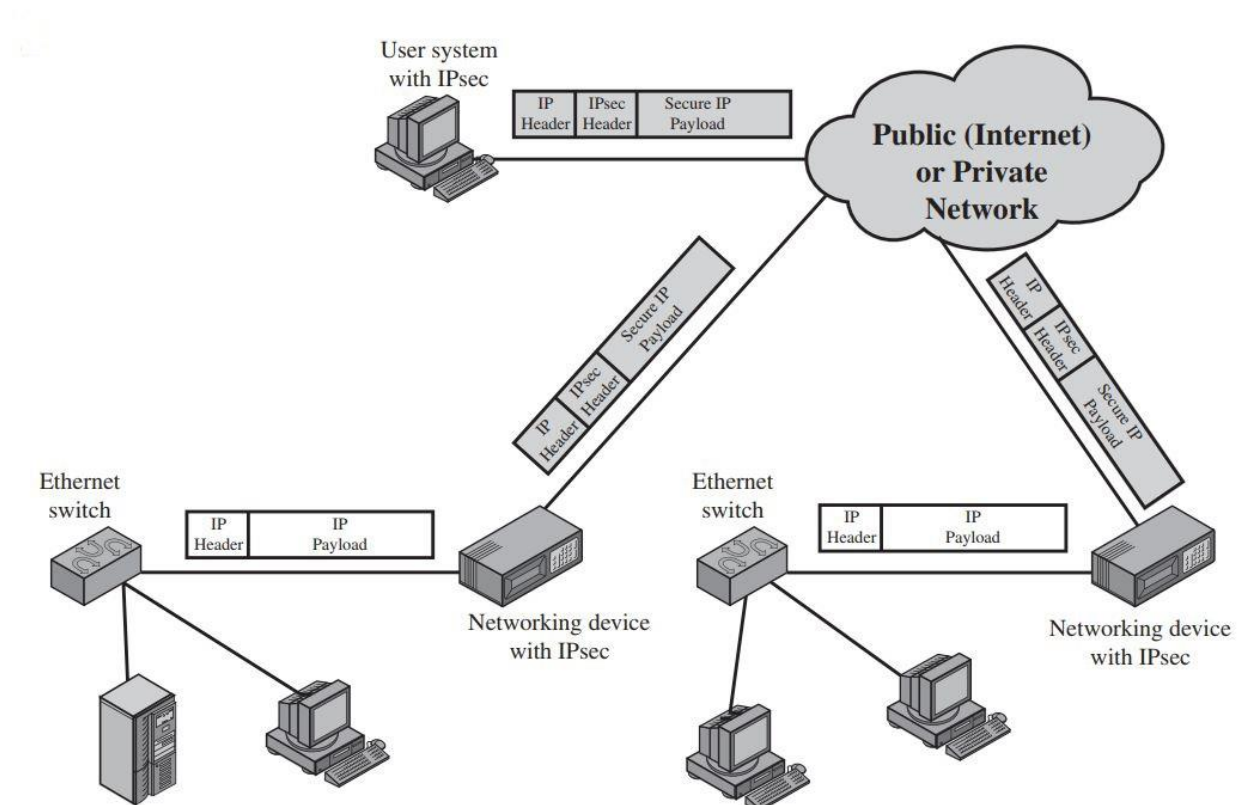


Figure: An IPsec Scenario

Three primary models for VPN architecture:

1. Gateway-to-Gateway

- Protects communications between two specific networks.
- Eg. An organization's main office network and a branch office network, or two business partners networks.

2. Host-to-gateway

- Protects communications between one or more individual hosts and a specific network belonging to an organization.
- Eg. Traveling employees to gain access to internal organizational services, such as the organization's e-mail and Web servers.

3. Host-to-Host

- Protects communications between two specific computers.
- Eg. Small number of users need to use or administer a remote system.

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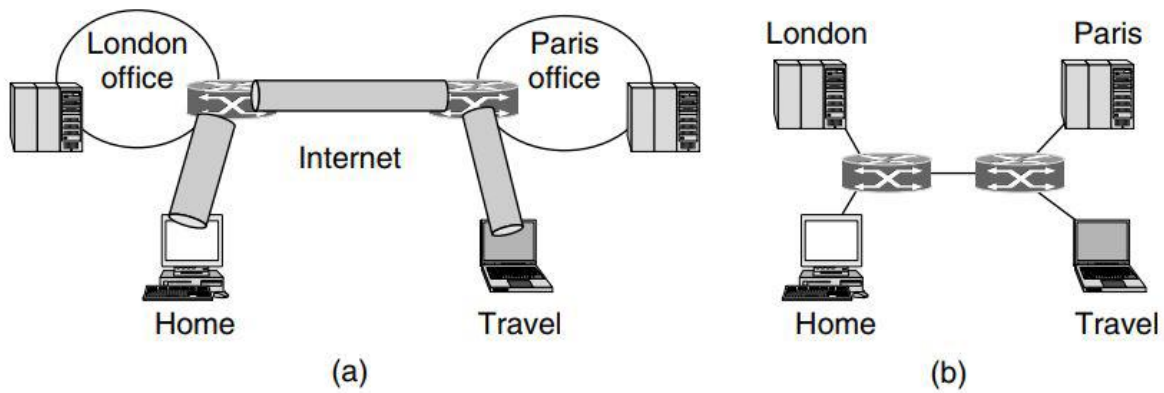


Figure: a) A VPN b) Topology as seen from the inside.